# Alireza Mahdizadeh

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## Research Interests

- **Reinforcement Learning & Continual Adaptation:** Experience Replay, Representation Learning, Catastrophic Forgetting, and Lifetime Tuning in RL Agents.
- **Memory & Attention in Autonomous Agents:** Structured Memory Management, Dynamic Attention Steering, and Multi-Agent Behavioral Stability.
- **Mathematical Foundations of AI:** Probabilistic Modeling, Optimization, and Differential Geometry.

## Education

**Isfahan University of Technology (IUT)** *Oct 2025 – Expected 2027*
*M.Sc. in Artificial Intelligence and Robotics* Isfahan, Iran

- **GPA:** 16.75 / 20.00 (Equivalent to **3.50 / 4.00**)
- **Thesis:** *Dynamic Steering of Internal Mechanisms and Memory Management in Large Language Models for Maintaining Agent Behavioral Stability in Multi-Agent Simulations*
- **Supervisor:** Dr. Hamidreza Hakim Davoodi

**University of Tehran (UT)** *Sep 2021 – May 2025*
*B.Sc. in Computer Engineering* Tehran, Iran

- **GPA:** 17.95 / 20.00 (Equivalent to **3.85 / 4.00**) — **Ranked 1st in Cohort** (for three consecutive years, 2022–2025)
- **Thesis:** *Using Deep Learning to Analyze Personality and Emotions in Text*

## Publications & Manuscripts

**Peer-Reviewed Journal Articles**

- M. Salarinoghabi, **A. Mahdizadeh**, and A. Cambraia Jr., "Return to the Reflection Symmetry of Plane Curves," *Bulletin of the Iranian Mathematical Society* (Springer), Vol. 52, No. 18, Mar. 2026.
  DOI: 10.1007/s41980-026-01036-z

**Manuscripts Under Review**

- **A. Mahdizadeh** and A. Hasan-Zadeh, "Curvilinear Reflection Through Tubular Neighborhoods: Metric Distortion, Singular Values, and the Role of Torsion," Submitted to *Results in Mathematics* (Springer), Manuscript ID: `RIMA-D-26-01982`, Oct. 2026.

**Books & Academic Translations**

- **Translator:** *"Machine Learning Handbook"* by Martha White. Translated the mathematical and theoretical notes into Persian to improve accessibility of core machine learning concepts for Persian-speaking researchers (2023).
- **Author:** *"Artificial Intelligence: The Last Achievement of Humanity"* (in Persian). Authored an analytical book exploring the philosophical perspectives, cognitive implications, and foundational paradigms of artificial intelligence (2023).

## Research Experience & Selected Technical Projects

**Master's Thesis Research: Memory & Attention Steering in Agents** *IUT | Oct 2025 – Present*

- Designing dynamic attention-steering mechanisms and structured memory architectures to mitigate behavioral drift and maintain long-horizon stability in interactive agents.
- Investigating how selective memory retrieval and internal representation control support continual adaptation in multi-agent environments.

**Reinforcement Learning & Probabilistic Modeling Implementations** *2024 – 2026*

- **Experience Replay Benchmark (`endpoint_replay_benchmark.py`):** Implemented and benchmarked experience replay sampling strategies in Python to evaluate sample efficiency, memory retention, and learning stability near trajectory endpoints in reinforcement learning tasks.

- **Statistical & Optimization Foundations in Code:** Developed an open-source repository of modular Python implementations covering maximum likelihood estimation (MLE), maximum a posteriori (MAP), regularization techniques, and gradient-based numerical optimization from scratch.

**Quantifying Gender Bias in Word Embeddings (Independent Research)** *2025*

- Conducted a statistical and geometric analysis of pre-trained GloVe word embeddings across 380+ occupational titles, measuring subspace projection biases against empirical U.S. Bureau of Labor Statistics workforce data.

**Bachelor's Thesis: Deep Learning for Emotion & Personality Detection** *UT | 2024 – 2025*

- Developed a multi-label NLP classification pipeline on the GoEmotions dataset using Word2Vec, GloVe, and Transformer representations across LSTM, GRU, BiLSTM, and BiGRU architectures (achieved micro-F1 of 48.9% and validation AUC of 0.6009).

## Teaching Experience

**Teaching Assistant** *Feb 2023 – Jun 2024*
*University of Tehran* Tehran, Iran

- **Data Structures and Algorithms** (under Dr. Ali Naghash Asadi) *Spring 2024 (Feb 2024 – Jun 2024)*
  Mentored undergraduate students on algorithm design, computational complexity analysis, graph algorithms, and C++ implementations.
- **Engineering Statistics & Probability** (under Dr. Atefeh Hasanzadeh) *Fall 2023 (Sep 2023 – Jan 2024)*
  Provided instruction on statistical estimation theory, probability distributions, random variables, and data interpretation techniques.
- **Engineering Mathematics** (under Dr. Atefeh Hasanzadeh) *Spring 2023 (Feb 2023 – Jun 2023)*
  Led problem-solving sessions, clarified advanced calculus, differential equations, and Fourier analysis concepts, and graded assignments.

## Honors & Awards

- **Ranked 1st in Cohort:** Achieved the highest cumulative GPA among all Computer Engineering entrants at the University of Tehran for three consecutive academic years (2022–2025).
- **Faculty of Engineering Best Student Award:** University of Tehran (2022–2023).
- **FOE Merit Scholarship Recipient:** Faculty of Engineering, University of Tehran.
- **National Data Science Competitions:** Ranked in the **99th** and **92nd percentiles** nationwide.

## Technical Skills & Academic Service

- **Programming & Frameworks:** Python, C++, PyTorch, Scikit-learn, NumPy, Pandas, vLLM, Ollama, MATLAB.
- **Tools & Environment:** Linux (Ubuntu), Git, LaTeX, Miniconda.
- **Academic Service:** Content Creator and Technical Writer for *BIT Student Scientific Journal*, University of Tehran.
- **Languages:** Persian (Native), English (Intermediate – Actively working on fluency).